BLOCKCHAIN · SMART CONTRACTS · E-VOTING

SmartVote

A Blockchain-Based Secure Electronic Voting System

Email-OTP authentication, wallet-ownership verification and on-chain tallying

The Problem

Centralised e-voting asks everyone to trust one operator and one database. Records can be altered unseen, and voters cannot check that their ballot was counted.

Our Solution

Every ballot is recorded and tallied by a Solidity smart contract on an Ethereum-compatible chain. Admins configure elections but can never cast, alter or delete a vote.

Four Layers of Voter Eligibility

1
E-mail OTP
Proves the voter controls an inbox

2
Admin approval
A human accepts the person into the electorate

3
Wallet proof
A signed one-time nonce proves key ownership

4
On-chain allow-list
The contract itself is the final gate

System Architecture

React Front-end
React 18, Vite, Tailwind, Ethers.js, MetaMask

→

Express API + MongoDB
JWT, bcrypt, rate limits. Stores no vote counts

→

Voting.sol Contract
Solidity 0.8.24, OpenZeppelin, Hardhat

Voters sign their own **vote()** in MetaMask. The back-end re-validates every reported transaction against the chain, and an event listener reconciles **VoteCast** events with the database.

68,351	102,551	47%	18 + 30+
gas per ballot (steady state)	gas for first ballot of an election	max saving from batch voter registration	contract and API tests, all passing

Security by Design

One vote per wallet, accepted only inside the election time window.

Tallies stay hidden until the election closes.

No name, e-mail or voter ID is stored on-chain.

Honest Limits and Future Work

Votes link a wallet to a candidate, so there is integrity and auditability but no ballot secrecy.

Next: commit–reveal, blind signatures and zero-knowledge proofs.

PROJECT TEAM

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